

FOCUS: Closed-Loop Attention Feedback for Efficient Vision-Language Understanding

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TL;DR Run the VLM only when the scene changes in a semantically meaningful way, then feed its own decoder attention back as the prior for the next call. Training-free.

74.8%

EgoSchema accuracy
uniform sampling: 74.6%

25.4

VLM calls per clip
uniform sampling: 55

57%

fewer captioning calls
at matched quality

1 Motivation

Applying VLMs to video remains expensive: a 30-fps video lasting one minute produces 1,800 frames, each adding hundreds of visual tokens to the model’s context. Where video arrives as a continuous stream, the full clip is never available in advance and the system must decide what to process as frames arrive. Existing VLM pipelines either assume access to the complete video or allocate compute uniformly across frames regardless of content.

Three existing families

Uniform temporal sub-sampling is blind to content, allocating the same compute budget to static segments as to brief, information-dense events. Token pruning and temporal compression lower per-call cost but still invoke the model on every sampled frame. Adaptive frame selection typically requires multiple selection rounds, an auxiliary retrieval model, or access to the full video, making it unsuitable for streaming.

Our work differs from all three in two ways

First, FOCUS reduces the *total number* of VLM calls through content-adaptive gating: it measures inter-frame dissimilarity in a VLM patch-embedding space rather than in pixel space, and triggers inference only when the scene changes meaningfully. Second, it closes the loop between perception and understanding: after each VLM call the decoder’s attention is fed back as a spatial prior for the next observation, without gradient updates or learned parameters.

2 Method

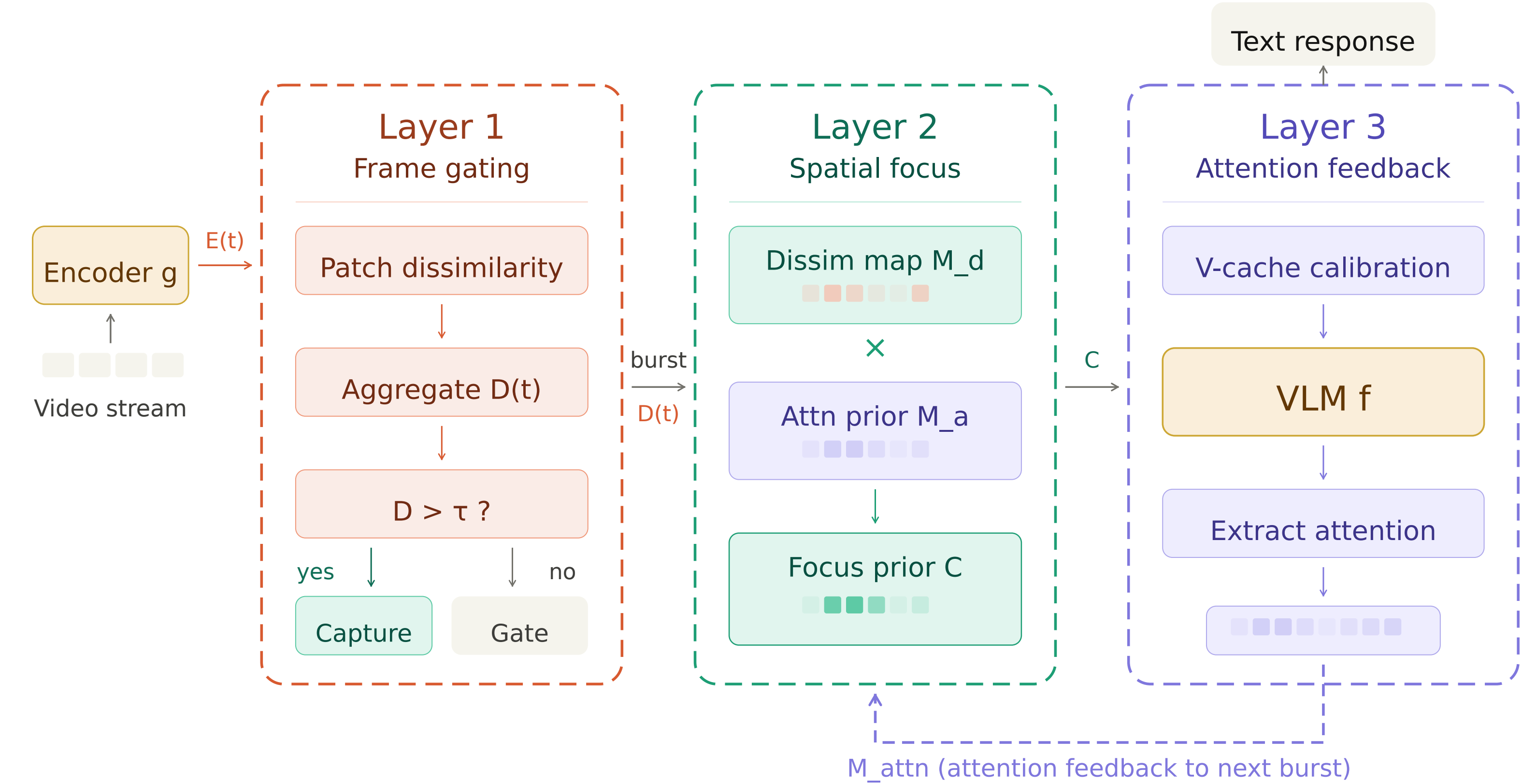


Figure 1: Architecture of the three-layer FOCUS system.

FOCUS asks: (1) should this frame be processed at all? (2) where in the frame should the VLM focus? (3) how can the VLM’s internal understanding improve future focus? Temporal gating and spatial attention have been explored independently; the contribution is closing the loop between them.

1 Dissimilarity in embedding space

FOCUS measures change in a VLM patch-embedding space rather than from pixel-level cues:

$$d_i^{(t)} = 1 - \cos(\mathbf{e}_i^{(t)}, \mathbf{e}_i^{(\text{ref})}), \quad D^{(t)} = \frac{1}{P} \sum_i d_i^{(t)}$$

Because dissimilarity is computed over learned semantic features, a meaningful change (a hand entering the scene) produces a large $D^{(t)}$ even if it occupies few pixels; camera shake may produce a small one.

2 Gating state machine

In CAPTURE the system collects a burst of B consecutive frames, giving the VLM a short temporal window to reason about an action, then transitions to WATCH, where frames are skipped. It returns to CAPTURE when $D^{(t)} > \tau$ or when a safety timeout of $N_{\max}$ skipped frames is reached.

3 Spatial prior construction

The per-patch dissimilarity is a byproduct of gating, but it also encodes *where* change occurred; max-pooled over the burst it gives $\mathbf{M}_{\text{dissim}}$. Not all change is task-relevant: a flickering monitor gives high dissimilarity without contributing to the task. We therefore fuse it with a top-down attention prior:

$$\mathbf{C} = \text{norm}(\mathbf{M}_{\text{dissim}} \odot (\varepsilon + \mathbf{M}_{\text{attn}}^V))$$

A region must score highly on *both* signals to receive a strong prior.

4 Value-cache scaling

Each vision token j receives a prior C_j and a per-step scale s_j , applied to the value cache at every autoregressive decoding step:

$$\mathbf{V}_j^{(\ell)} \leftarrow s_j \cdot \mathbf{V}_j^{(\ell)}, \quad s_j = 1 + (\alpha_f - 1) C_j$$

Factors compound: at our 100-token cap a peak-prior token is scaled $\approx 131\times$ and a fully off-prior token $\approx 0.017\times$. Keys and queries are untouched, so attention *weights* are unchanged and no token leaves the context.

5 Closed-loop attention feedback

Most frame selection systems are purely feed-forward: the selector never receives feedback from the model that consumes its output. After each burst a single additional forward pass yields attention tensors; the last $N_a=4$ layers are max-pooled across heads, averaged across layers and folded back onto the patch grid as $\mathbf{M}_{\text{attn}}$.

$$\tilde{\mathbf{A}} = \frac{1}{N_a} \sum_{\ell=L-N_a}^{L-1} \max_h \mathbf{A}^{(\ell,h)}$$

$\mathbf{M}_{\text{attn}}$ is retained across bursts, so the VLM’s understanding from burst n shapes the prior of burst $n+1$; the first burst runs without calibration, an orienting phase.

3 Experimental setup

FOCUS uses the Qwen2-VL-2B-Instruct vision encoder for gating and Qwen3-VL-8B-Instruct as the inference VLM; the same backbone is used for *every* method in every table. One configuration for every experiment: $\tau = 0.4$, $B = 32$, $\alpha_f = 1.05$, $\beta = 0.04$, greedy decoding, 100-token cap, all fixed before any evaluation run on a disjoint development set. Baselines under one backbone: **uniform sampling**, **DyCoke**, **BOLT**, and **FOCUS (gating only)**, isolating gating from everything downstream.

Efficiency metric

ρ is the ratio of total VLM inference time to video duration; ρ_{e2e} also charges the gating encoder, whose cost $t_{\text{enc}} = 34$ ms per frame FOCUS pays on every arriving frame while uniform sampling and DyCoke do not:

$$\rho = \frac{\sum_n t_{\text{vlm}}^{(n)}}{T_{\text{video}}}, \quad \rho_{\text{e2e}} = \frac{\sum_n t_{\text{vlm}}^{(n)} + t_{\text{enc}} N_{\text{frames}}}{T_{\text{video}}}$$

4 Results

4.1 EgoSchema under a matched backbone

All methods share the Qwen3-VL-8B backbone at a matched 3.3 s cadence, in the *streaming* setting: frames arrive in order, with no future frames and no pre-specified query.

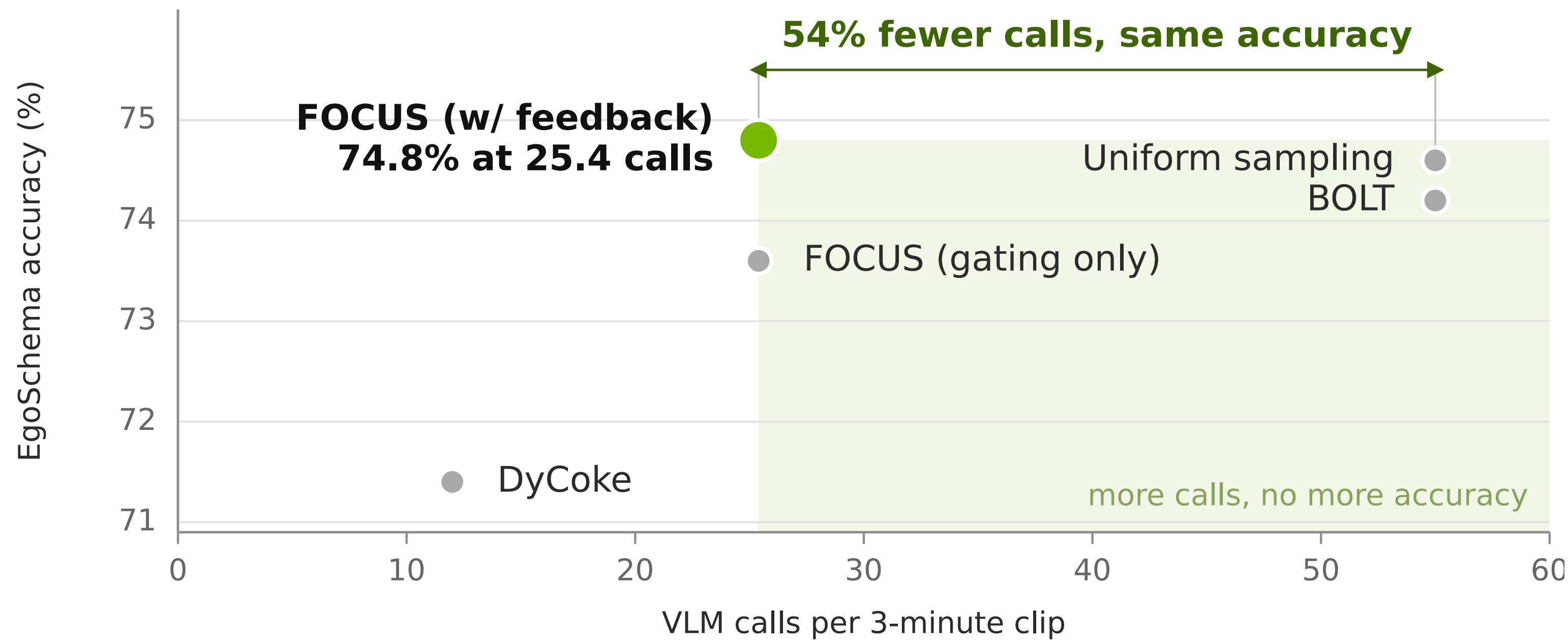


Figure 2: EgoSchema subset accuracy (500 videos, 5-way multiple choice) against mean VLM invocations per three-minute clip, mean over five runs. The two FOCUS points issue identical calls by construction: gating alone determines how many bursts fire.

FOCUS with feedback matches the strongest baselines’ accuracy at under half their call budget (25.4 vs. 55). We claim no accuracy *win*: one video is 0.2 points on 500 items, so the gaps to BOLT and to uniform are three videos and one. DyCoke’s 3.4-point deficit is the price of compressing each call rather than deciding whether to make it.

4.2 Egocentric video captioning

Method	Actor	Activ.	Obj.	NoHal.	Overall	ρ	ρ_{e2e}	Calls
Uniform Sampling	4.96	4.12	4.00	2.16	3.81	4.14	4.14	516
DyCoke	4.80	3.88	4.04	2.52	3.81	3.07	3.07	263
BOLT	4.76	4.12	4.08	2.68	3.91	0.87	≥ 0.87	263
FOCUS (gating only)	5.00	4.00	4.12	2.44	3.89	0.57	1.34	222
FOCUS (w/ Feedback)	5.00	3.96	4.12	2.60	3.92	0.59	1.36	222

Table 1: Captioning quality (GPT-5.2, 1–5) and efficiency on 25 egocentric videos; lower ρ is better. Same backbone throughout, but not decoding-matched: DyCoke and BOLT use sampled decoding and a 200-frame interval.

The headline reading is cost, not quality. FOCUS uses 57% fewer VLM calls than uniform sampling and 16% fewer than BOLT and DyCoke, at $\rho = 0.59$; its overall quality (3.92) is statistically indistinguishable from BOLT’s (3.91), paired Wilcoxon $p = .895$. The claim is the same quality at two-thirds of BOLT’s compute ratio, under a streaming constraint BOLT does not satisfy. ρ_{e2e} charges the gating encoder in full; that term is a constant 0.77, so pipelining gives 0.88.

5 Ablations

Configuration	α_f	β	MCQ	Judge	Obj.
(a) Gating only	-	-	76.0%	3.81	3.76
(b) + amplification	1.05	-	76.0%	3.84	3.96
(c) + suppression	1.05	0.04	78.0%	3.81	3.92

Table 2: Component ablation, 50 EgoSchema videos. Row (a) is FOCUS (gating only), row (c) is FOCUS (w/ Feedback) as deployed.

Gating accounts for the entire reduction in calls; calibration supplies the quality steering at the cost of one extra forward pass per burst. The clearest effect is on object grounding, rising from 3.76 to 3.96 once amplification is enabled; at $n=50$ the deltas sit inside the cohort’s confidence band. τ trades compute against marginal quality: judge quality is flat at 3.81–3.88 from $\tau = 0.2$ to 0.5 while ρ falls threefold. We choose $\tau = 0.4$ as the knee, explicitly *not* the accuracy maximum.

6 Spatial prior evolution

The spatial prior tends to concentrate on regions of activity, such as a robot arm or the object it manipulates, without explicit region supervision.

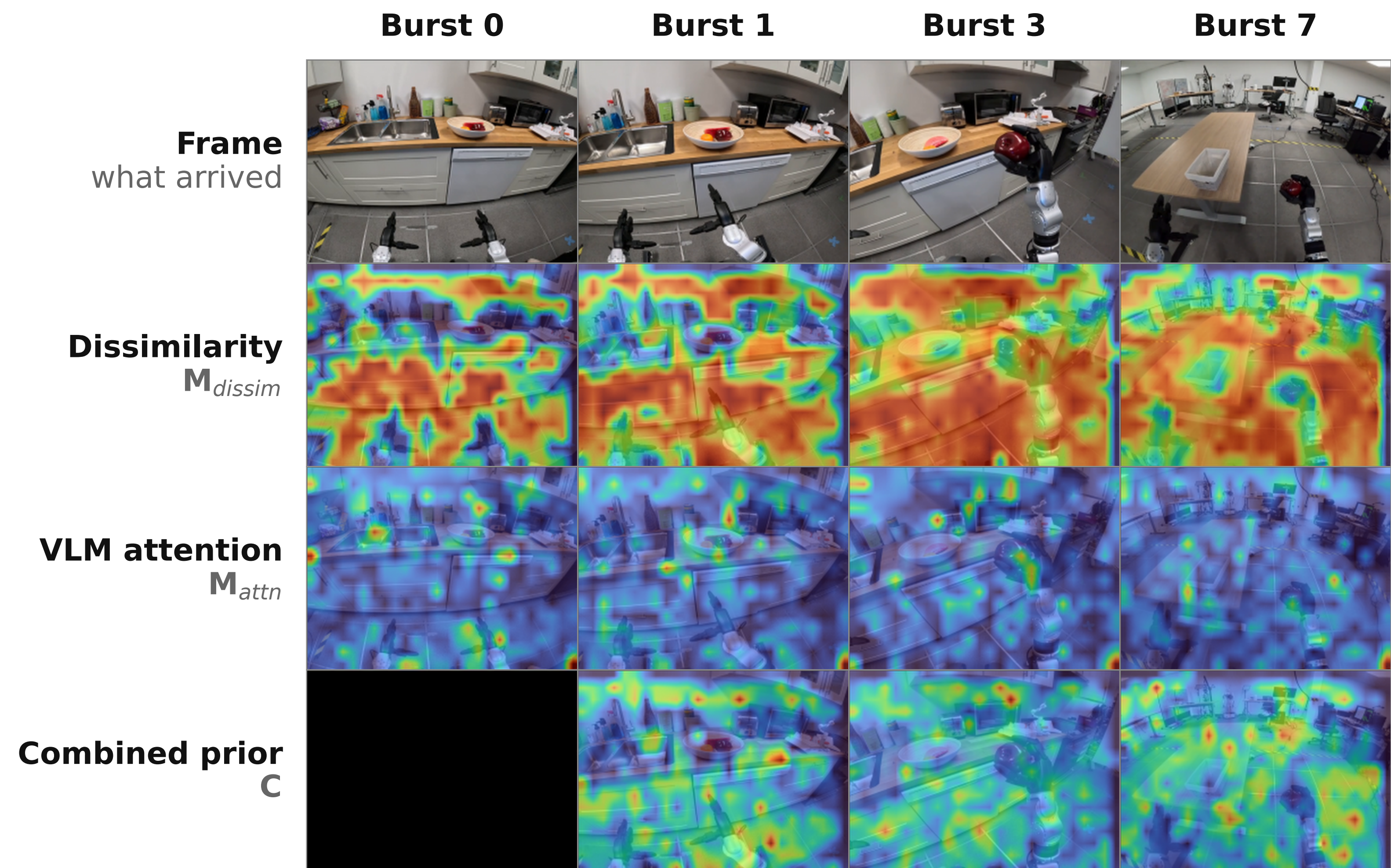


Figure 3: Spatial prior evolution across bursts (robotic teleoperation). Rows are a representative frame, the dissimilarity map, the decoder attention after generation, and the combined prior, absent at Burst 0, the orienting phase.

7 Qualitative case studies: FOCUS without attention feedback against full FOCUS, on EPIC-Kitchens



Figure 4. Frames from the soup-stirring clip (EPIC-Kitchens, 15 s). Both variants select the same two bursts and give identical first-burst descriptions, since the orienting phase has no prior; their second-burst descriptions diverge.

Without feedback

| “placed the metal lid onto the right-hand burner of the stove, covering the empty pot.”

With feedback

| “placed the metal lid onto the pot on the right side of the stove, covering it. The action has shifted from stirring or preparing to actively covering the pot, likely to let its contents simmer or cook.”

Attention feedback corrects spatial hallucinations. Without feedback the model hallucinates an “empty pot” and misidentifies the lid’s destination as a burner rather than the pot. With feedback, the prior extracted from Burst 1, where the model attended heavily to the pot and its contents, persists into Burst 2 and anchors the description to the correct object.



Figure 5. Frames from the sauce-stirring clip (EPIC-Kitchens, 25 s). Third-burst descriptions, by which point the spatial prior has converged on the spoon–bowl–pot region.

Without feedback

| “The person has now lifted the spoon ... poised to transfer the sauce. The spoon is suspended above the bowl, indicating the action of pouring is about to begin ... The sauce in the pot appears to be thick and reddish-brown, and the bowl is empty and ready to receive the contents.” (5.5 s inference)

With feedback

| “The person has now lifted the spoon ... poised to transfer the sauce.” (3.2 s inference)

Feedback produces more focused descriptions. By the third burst the spatial prior has converged on the spoon–bowl–pot region, so the model does not re-describe the scene layout or the sauce’s appearance; the description is more concise and 41% faster to generate. Uniform sampling on the same clip produces 15 bursts totalling 233 s of VLM compute ($\rho = 9.3\times$), 13 of them opening with the same phrase; FOCUS captures 3 bursts in 11 s ($\rho = 0.44\times$), each a distinct action transition.